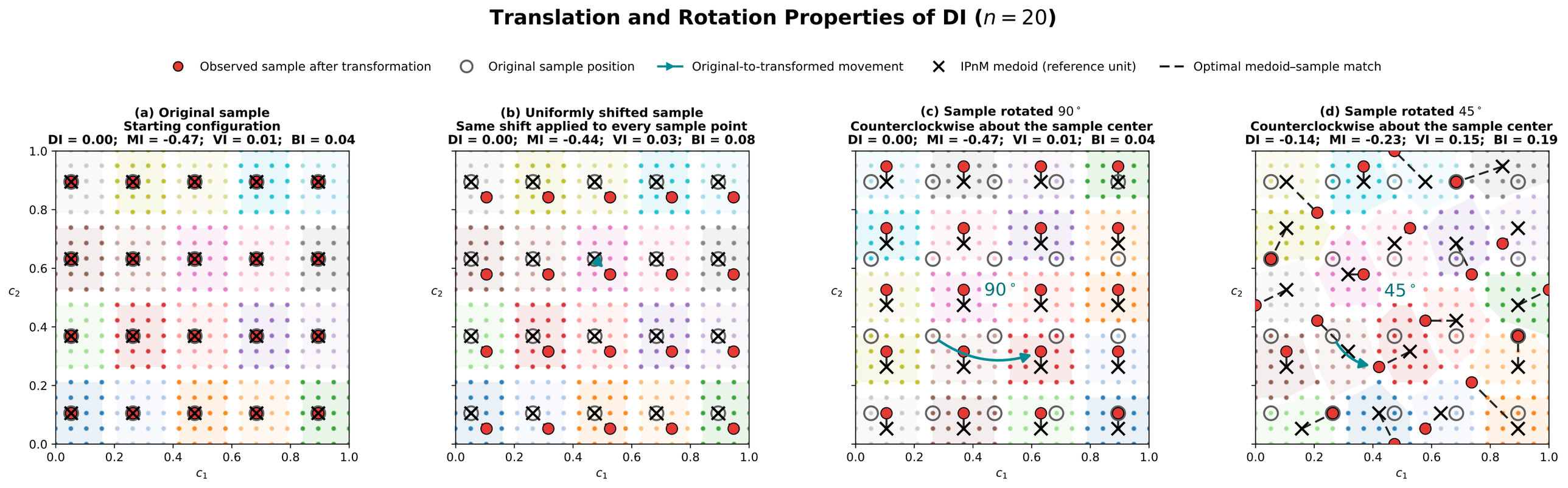

Translation and Rotation Properties of DI ($n = 20$)
Observed sample after transformation
Original sample position
Original-to-transformed movement
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Original sample
Starting configuration
DI = 0.00; MI = -0.47; VI = 0.01; BI = 0.04
(b) Uniformly shifted sample
Same shift applied to every sample point
DI = 0.00; MI = -0.44; VI = 0.03; BI = 0.08
(c) Sample rotated 90°
Counterclockwise about the sample center
DI = 0.00; MI = -0.47; VI = 0.01; BI = 0.04
90°
(d) Sample rotated 45°
Counterclockwise about the sample center
DI = -0.14; MI = -0.23; VI = 0.15; BI = 0.19
45°
c_1
c_2
0.0
0.2
0.4
0.6
0.8
1.0